

A Parallel Unstructured Finite-Volume Solver for Two-Dimensional Compressible Navier–Stokes Benchmarks

cf2d submission team

August 30, 2026

Abstract

This report describes `cf2d`, an original C++17/MPI cell-centered finite-volume solver for the two-dimensional compressible Navier–Stokes equations. The solver reads CGNS meshes, partitions a cell graph with METIS, exchanges only neighbor ghosts during iterations, reconstructs primitive variables to second order, and advances steady problems with an implicit pseudo-time inexact-Newton symmetric Gauss–Seidel relaxation. The transient cylinder case uses a BDF2 outer physical-time loop with inner nonlinear relaxation. Seven steady cases were advanced to the residual and force plateau defined by the solver. The transient Reynolds-200 case was run with corrected per-step limiter refresh to the full supplied horizon and finished in a statistically periodic vortex-street state. The report documents equations, discretization, boundary conditions, MPI strategy, verification, visualizations, and known limitations. The most important limitation is the under-predicted drag and nonzero lift obtained for the Reynolds-20 cylinder on the supplied mesh; this is discussed rather than hidden.

1 Introduction

The benchmark requires an original unstructured compressible Navier–Stokes solver and complete results for six zero-angle-of-attack NACA0012 cases and two circular-cylinder cases. The implementation is in `solver/src`. The main design goals were strict conservation, local mesh distribution, a genuinely implicit nonlinear relaxation, and reproducible output artifacts. The same executable handles steady and transient cases; case differences come from the JSON input and documented command-line options.

2 Governing Equations and Nondimensionalization

The two supplied meshes are NACA0012 (15,682 nodes, 20,816 cells) and cylinder (9,995 nodes, 10,185 cells). Their tags are `WALL` and `FAR`. table 1 summarizes the production controls used by the final runs; table 2 compares the supplied and CLI-used CFL and step limits, and the exact invocations are listed in `report/run_manifest.md`.

Rusanov is used in all rows except the Mach-0.8 inviscid airfoil and the Reynolds-200 transient case, for which the Roe flux with entropy fix is used.

The conservative state and governing system are

$$\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho E \end{bmatrix}, \quad \frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot (\mathbf{F}^i - \mathbf{F}^v) = \mathbf{0}. \quad (1)$$

Table 1: Case groups, boundary tags, and production numerical controls. Steady values are pseudo-time controls; the Reynolds-200 row is a physical-time control set.

Case group	Boundary tags	M_∞	Re	CFL_0	$CFL_{\max}$	Ramp	Inner	Target
NACA M0.15 inviscid	WALL/FAR	0.15	–	1	5	2000	[3,50]	0.01
NACA M0.8 inviscid	WALL/FAR	0.80	–	1	10	3000	[3,50]	0.01
NACA M2.0 inviscid	WALL/FAR	2.00	–	1	5	5000	[3,80]	0.01
NACA M0.15 laminar	WALL/FAR	0.15	5000	1	10	5000	[3,80]	0.01
NACA M0.8 laminar	WALL/FAR	0.80	5000	1	10	5000	[3,80]	0.01
NACA M2.0 laminar	WALL/FAR	2.00	5000	1	10	7000	[3,100]	0.01
Cylinder Re20	WALL/FAR	0.10	20	5	10	3000	[3,80]	0.01
Cylinder Re200	WALL/FAR	0.10	200	1	1	0	[5,1000]	10^{-3}

Table 2: Supplied and production pseudo-time CFL and step limits. The production invocations use the stricter CFL caps and step limits shown in the right-hand column; the Reynolds-200 transient controls remain as supplied.

Case	Supplied $CFL_0/CFL_{\max}/\text{ramp}/\text{max steps}$	Used $CFL_0/CFL_{\max}/\text{max steps}$
NACA M0.15 inviscid	1/100/2000/20000	1/5/1500
NACA M0.8 inviscid	1/100/3000/30000	1/10/8000
NACA M2.0 inviscid	0.2/50/5000/40000	1/5/2000
NACA M0.15 laminar	1/100/5000/40000	1/10/12000
NACA M0.8 laminar	0.5/80/5000/40000	1/10/12000
NACA M2.0 laminar	0.2/50/7000/50000	1/10/12000
Cylinder Re20	1/100/3000/30000	5/10/12000

The Cartesian inviscid fluxes are

$$\mathbf{F}^i = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho uv \\ u(\rho E + p) \end{bmatrix}, \quad \mathbf{G}^i = \begin{bmatrix} \rho v \\ \rho uv \\ \rho v^2 + p \\ v(\rho E + p) \end{bmatrix}. \quad (2)$$

For a calorically perfect gas,

$$p = (\gamma - 1)\rho e, \quad E = e + \frac{u^2 + v^2}{2}, \quad a^2 = \frac{\gamma p}{\rho}. \quad (3)$$

The Newtonian stress is

$$\boldsymbol{\tau} = 2\mu\mathbf{S} - \frac{2}{3}\mu(\nabla \cdot \mathbf{u})\mathbf{I}, \quad \mathbf{S} = \frac{1}{2}(\nabla \mathbf{u} + \nabla \mathbf{u}^T), \quad (4)$$

and the heat flux is $\mathbf{q} = -k\nabla T$ with $k = \mu c_p / \text{Pr}$. Constant viscosity is set from the case Reynolds number, $\mu = \rho_\infty U_\infty L_{\text{ref}} / \text{Re}$.

The supplied cases use $\rho_\infty = 1$, $U_\infty = 1$, and $L_{\text{ref}} = 1$; hence the dynamic pressure is $q_\infty = \rho_\infty U_\infty^2 / 2 = 1/2$. The force coefficients are

$$C_D = \frac{D}{q_\infty A_{\text{ref}}}, \quad C_L = \frac{L}{q_\infty A_{\text{ref}}}, \quad C_{m,z} = \frac{M_z}{q_\infty A_{\text{ref}} L_{\text{ref}}}. \quad (5)$$

The `cd` and `cl` columns in `forces.csv` are coefficients. For historical transparency, the four component columns retain the dimensional integrated force units; the report tables divide these columns by $q_\infty A_{\text{ref}}$. The reported $C_{m,z}$ is the pressure moment about the case moment center; viscous moment is not included.

3 Meshes, Inputs, and Boundary Families

CGNS zones are read in physical coordinates, merged cell by cell, and oriented counterclockwise. Near-duplicate coordinates are removed with a 10^{-10} quantization tolerance. Face connectivity is generated once from cell polygons. Zone boundary families named `WALL` and `FAR` are mapped to no-slip adiabatic or farfield conditions for cylinder cases and to slip or farfield conditions for inviscid NACA cases. Cell volumes and centroids use polygon formulas. The NACA0012 mesh has 15,682 nodes, 20,816 cells, and 36,498 faces; the cylinder mesh has 9,995 nodes, 10,185 cells, and 20,180 faces.

Small sliver cells are agglomerated into the neighbor across their longest shared edge when their volume is below 10^{-6} times the median cell volume. If merging occurs, final cell geometry and face topology are built from the merged polygons.

4 Spatial Discretization

For a face f with area A_f and unit normal $\mathbf{n}$ oriented from the left cell to the right cell, the residual of the left cell receives $\mathbf{F}(\mathbf{U}_L, \mathbf{U}_R) \cdot \mathbf{n} A_f$ and the right cell receives the negative. Ghost conservative states are synchronized before reconstruction and residual assembly. Viscous faces use cell-centered primitive gradients, directional correction of the interface gradient, and averaged face states.

4.1 Second-Order Reconstruction and Limiter

Green–Gauss gradients of ρ , u , v , p , and T are computed for every cell. Face states are piecewise linear,

$$\mathbf{w}_{L/R} = \mathbf{w}_c + \phi_c \nabla \mathbf{w}_c \cdot (\mathbf{x}_f - \mathbf{x}_c), \quad (6)$$

where $\mathbf{w} = [\rho, u, v, p, T]$. A Barth–Jespersen limiter provides the single coefficient ϕ_c from cell and face-neighbor extrema. If a reconstructed state is nonphysical, the face state falls back to the cell average; the update also restores the pre-update state when necessary. The steady solver adaptively freezes the limiter field only after limiter chatter blocks the inner relaxation. In transient iterations the limiter field is refreshed at the start of every physical step and any freeze applies only to the remainder of that step.

4.2 Inviscid and Viscous Fluxes

For the inviscid part of the face flux, the default choice in viscous configurations is Rusanov/local Lax–Friedrichs:

$$\hat{\mathbf{F}} \cdot \mathbf{n} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) \cdot \mathbf{n} - \frac{1}{2}s_{\max}^+(\mathbf{U}_R - \mathbf{U}_L). \quad (7)$$

The total face residual is the sum of this inviscid Riemann flux and the separately assembled central viscous flux. The Roe flux with a Harten-type entropy fix ($\epsilon = 0.05$) is available and was used for the Mach-0.8 inviscid airfoil and the Reynolds-200 transient case. Viscous tractions use the Newtonian stress tensor and Fourier heat flux. At no-slip walls, the wall state has $u = v = 0$, the adiabatic condition sets normal temperature flux to zero, and skin friction is the tangential component of wall shear rather than the full normal traction.

5 Boundary Conditions

Farfield states are reconstructed from a characteristic farfield ghost state based on freestream Mach, angle of attack, density, and pressure. Inviscid slip walls mirror the normal velocity while preserving pressure and tangential velocity. No-slip adiabatic walls enforce zero wall velocity and zero normal heat flux. `surface.csv` reports boundary states: no-slip rows have $u = v = M = 0$, and slip rows have near-zero normal velocity. The pressure and skin-friction coefficients are evaluated at these boundary states.

6 Implicit and Transient Time Integration

Steady cases use local pseudo-time stepping. The local timestep contains both convective and viscous spectral-radius estimates. Each pseudo step performs up to the case maximum inner iterations of a symmetric Gauss–Seidel sweep with a spectral-radius Jacobian and fresh forward/backward sweeps. Gradients and limiter values are treated as frozen coefficients inside an inner solve and refreshed once per accepted step. Convergence requires the case residual target, a stationary force tail, and no non-finite states. For final steady runs the maximum tail deviations of C_L and C_D were required to be below 5×10^{-3} .

The transient cylinder case uses BDF2 in physical time:

$$\frac{3\mathbf{U}_i^{n+1} - 4\mathbf{U}_i^n + \mathbf{U}_i^{n-1}}{2\Delta t} + \mathbf{R}_i(\mathbf{U}^{n+1}) = \mathbf{0}, \quad \Delta t = 0.01. \quad (8)$$

The physical-time states $\mathbf{U}^n$ and $\mathbf{U}^{n-1}$ are frozen during all inner iterations and are shifted only after the full transient residual reaches the inner target. The production run uses $\Delta t = 0.01$, final time 300, inner bounds $[5, 1000]$, and inner residual ratio 10^{-3} . A first-order BDF1 term is used only for the very first step after a fresh start.

7 MPI Parallelization

Rank zero reads the global mesh, builds a cell-adjacency graph, and calls METIS k-way partitioning. The partition is cached for identical mesh and rank pairs, then local owned cells, ghosts, faces, and neighbor communication plans are scattered. Each rank stores only its local mesh and conservative state. Ghost states, gradients, and limiter values are exchanged by neighbor-scoped `MPI_Isend/Irecv` calls with deterministic tags. Residual and force norms use global MPI reductions. table 3 summarizes representative np=8 partitions.

Table 3: Representative np=8 partition diagnostics. Send and receive totals are summed over ranks; the submitted CSVs also contain per-rank neighbor lists.

Case	Owned cells by rank				Ghost range		Neighbors	Cut	Send/Recv
	min	max	mean	max/mean	min	max			
NACA0012	2422	2977	2602.0	1.14	73	189	2/6	435	860/860
Cylinder	1234	1299	1273.1	1.02	92	145	3/6	500	933/933

8 Output, Reproducibility, and Sanity Checks

The solver is built with

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=Release \
  -DCFD_EXTERNALS_ROOT=/opt/external/cfd_externals/install
cmake --build build
```

and each run uses `scripts/run_case.sh`. Every case directory contains `metadata.json`, `partition_diagnostics.residuals.csv`, `forces.csv`, `surface.csv`, `field_final.vtk`, per-rank restart files, `stdout.log`, and `run_status.json`. `tools/plot_results.py`, `tools/analyze.py`, `tools/make_manifests.py`, `tools/make_report_tables.py`, and `tools/rank_compare.py` regenerate figures, checks, manifests, tables, and rank comparisons from these submitted data files. The final machine-readable checks are `report/sanity_checks.json`; figure-to-source mapping is `report/figure_manifest.csv`; and exact commands are `report/run_manifest.md`. The final analysis reports positive density and pressure, near-zero reported wall velocity for no-slip rows, nonzero wall C_p , positive cylinder drag, and nonzero unsteady lift when the Reynolds-200 case reaches its shedding state.

9 Results

9.1 Run Status and Force Summary

table 4 lists completion status, residual reduction, and wall time. Residual reduction is measured from the first global residual norm formed by the solver. table 5 lists mean coefficients over the last 20 accepted steady steps, or the statistically settled post-transient tail for the Reynolds-200 case.

table 6 reports the actual inner-solve statistics. Many steady pseudo steps use the maximum inner count because the per-step 10^{-2} inner residual ratio is not reached; the outer residual target plus the tightened force plateau is therefore the operative convergence criterion.

Table 4: Run status for production cases. Physical time is zero for steady cases.

Case	Ranks	Step	t	Orders	Wall (s)	Status
cylinder_m010_laminar_re20	8	794	0.00	6.33	176	converged
cylinder_m010_laminar_re200	8	30001	300.01	4.88	6949	statistically_periodic
naca0012_m015_inviscid	8	1034	0.00	5.47	349	converged
naca0012_m015_laminar_re5000	8	861	0.00	6.01	549	converged
naca0012_m080_inviscid	8	758	0.00	4.26	410	converged
naca0012_m080_laminar_re5000	8	601	0.00	4.05	515	converged
naca0012_m200_inviscid	8	33	0.00	3.24	25	converged
naca0012_m200_laminar_re5000	8	127	0.00	3.64	140	converged

Table 5: Mean force coefficients and normalized pressure/viscous components.

Case	C_D	C_L	$C_{m,z}$	$C_{D,p}$	$C_{D,v}$	$C_{L,p}$	$C_{L,v}$
cylinder_m010_laminar_re20	0.84691	-0.38968	0.00000	0.68352	0.16338	-0.40325	0.01357
cylinder_m010_laminar_re200	1.12646	-0.00147	0.00000	1.01002	0.11643	-0.00292	0.00146
naca0012_m015_inviscid	0.00595	-0.00260	0.00086	0.00595	0.00000	-0.00260	0.00000
naca0012_m015_laminar_re5000	0.05009	0.01133	-0.00275	0.01587	0.03422	0.01105	0.00029
naca0012_m080_inviscid	0.01063	-0.00021	0.00038	0.01063	0.00000	-0.00021	0.00000
naca0012_m080_laminar_re5000	0.07589	-0.01234	-0.00697	0.04017	0.03572	-0.01221	-0.00013
naca0012_m200_inviscid	0.09400	-0.00126	0.00026	0.09400	0.00000	-0.00126	0.00000
naca0012_m200_laminar_re5000	0.06549	0.00042	0.00025	0.04762	0.01787	0.00022	0.00019

9.2 NACA0012 Cases

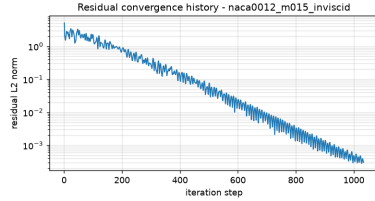
Figure 1 and fig. 2 show the convergence and force histories; fig. 3 and fig. 4 show wall distributions. fig. 5 and fig. 6 give the Mach-0.15 inviscid detail; fig. 7 and fig. 8 show its transonic and supersonic counterparts, while fig. 9, fig. 10, and fig. 11 show the laminar fields. At zero angle of attack the computed lift magnitudes are much smaller than drag, as expected by symmetry, although not exactly zero because the supplied unstructured mesh is not reflection symmetric. The low-speed inviscid drag is $O(10^{-2})$ or smaller, the transonic case shows a shock on the suction and pressure surfaces, and the supersonic case has sharper wave structure. Laminar cases show finite wall shear and distinct `surface.csv` pressure and friction distributions.

9.3 Cylinder at Reynolds 20

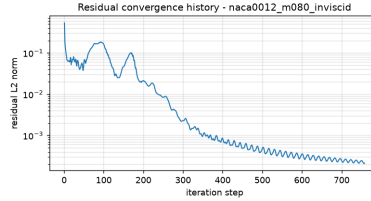
fig. 14 shows the convergence and force plateau. figs. 12, 13 and 15 show the steady cylinder result. The solver reaches a stable residual/force plateau with positive drag, but the computed drag $C_D \approx 0.847$ is below the commonly reported incompressible Reynolds-20 value near 2, and $C_L \approx -0.390$ rather than zero. A reflection check of the supplied mesh does not find an exact mirror node for most nodes, and the wall C_p distribution is asymmetric. The cylinder rank comparison

Table 6: Inner-solve statistics from `metadata.json`. Requested and observed min/max iterations, misses of the per-step target, converged-step fraction, and final inner residual ratio are listed.

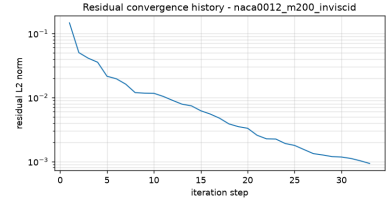
Case	Req. min	Req. max	Typical	Obs. min	Obs. max	Misses	Fraction	Last ratio
cylinder_m010_laminar_re20	3	80	80	80	80	794	0.000	0.707
cylinder_m010_laminar_re200	5	1000	246	194	278	0	1.000	0.000994
naca0012_m015_inviscid	3	50	50	50	50	1034	0.000	0.931
naca0012_m015_laminar_re5000	3	80	80	80	80	861	0.000	0.871
naca0012_m080_inviscid	3	50	50	50	50	758	0.000	0.999
naca0012_m080_laminar_re5000	3	80	80	80	80	601	0.000	1.01
naca0012_m200_inviscid	3	80	80	80	80	33	0.000	0.296
naca0012_m200_laminar_re5000	3	100	100	100	100	127	0.000	0.683



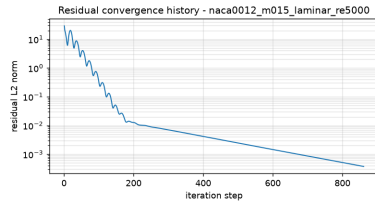
(a) M0.15 inviscid.



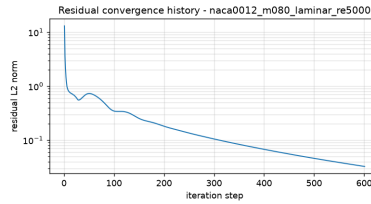
(b) M0.80 inviscid.



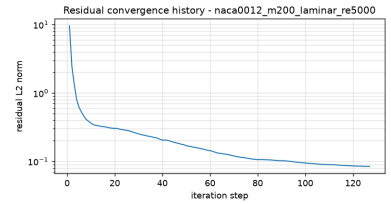
(c) M2.0 inviscid.



(d) M0.15 laminar.



(e) M0.80 laminar.



(f) M2.0 laminar.

Figure 1: Global L2 residual histories for all NACA0012 cases, plotted from `residuals.csv`.

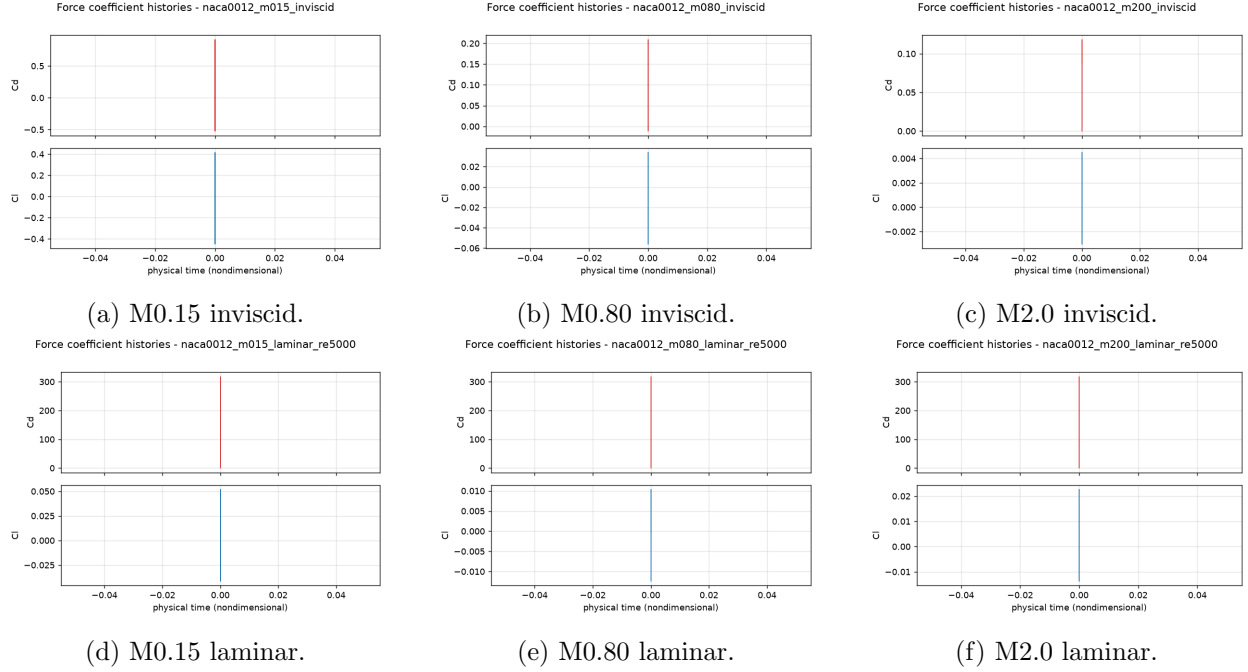


Figure 2: Lift and drag coefficient histories for all NACA0012 cases, plotted from `forces.csv`.

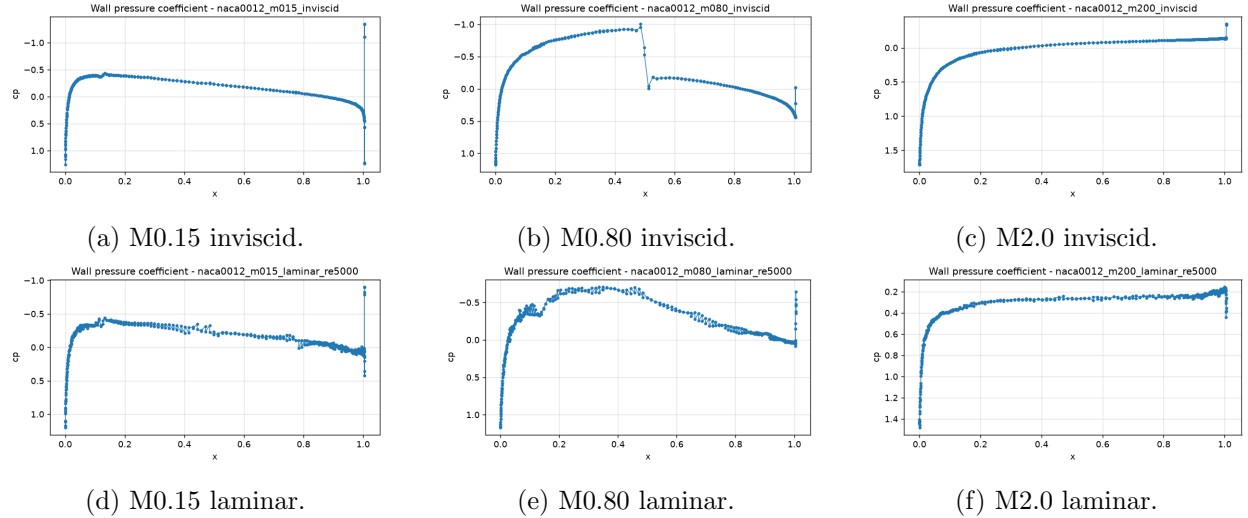


Figure 3: Wall pressure coefficient distributions for all NACA0012 cases, reported at the body boundary.

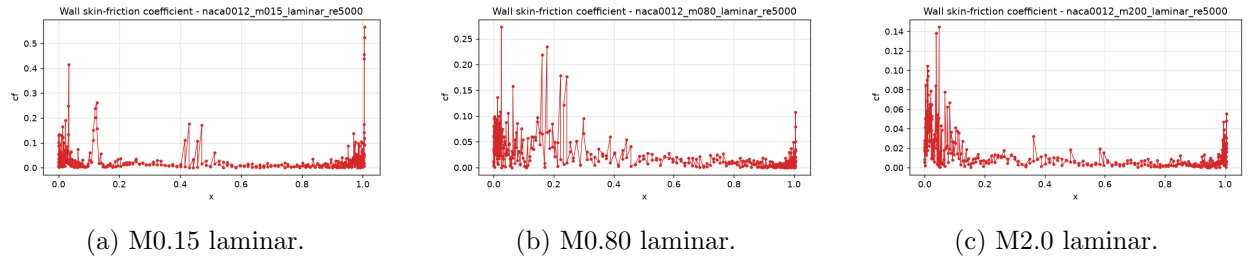


Figure 4: Wall skin-friction coefficient distributions for the laminar NACA0012 cases.

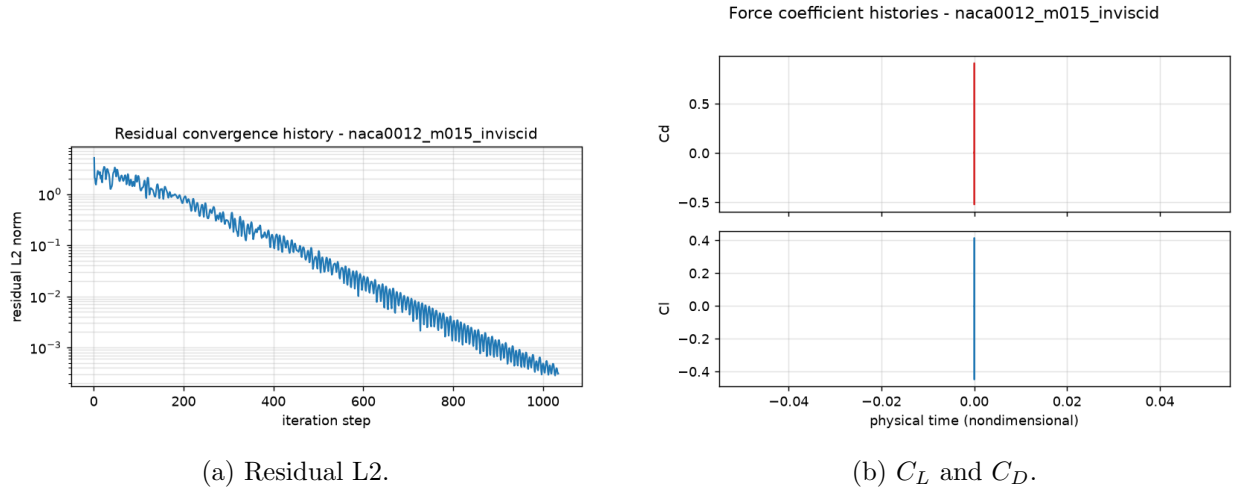


Figure 5: Mach-0.15 inviscid NACA0012 histories from `residuals.csv` and `forces.csv`.

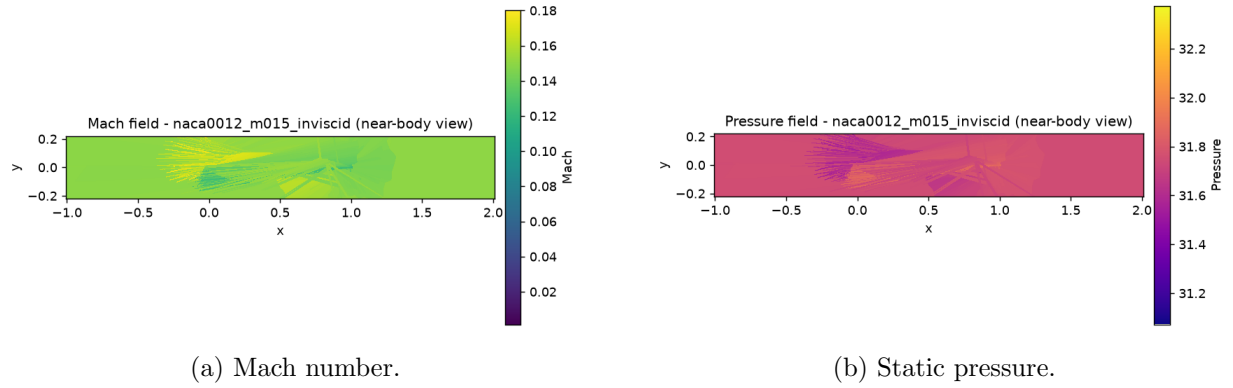


Figure 6: Near-body Mach and pressure contours for the Mach-0.15 inviscid airfoil.

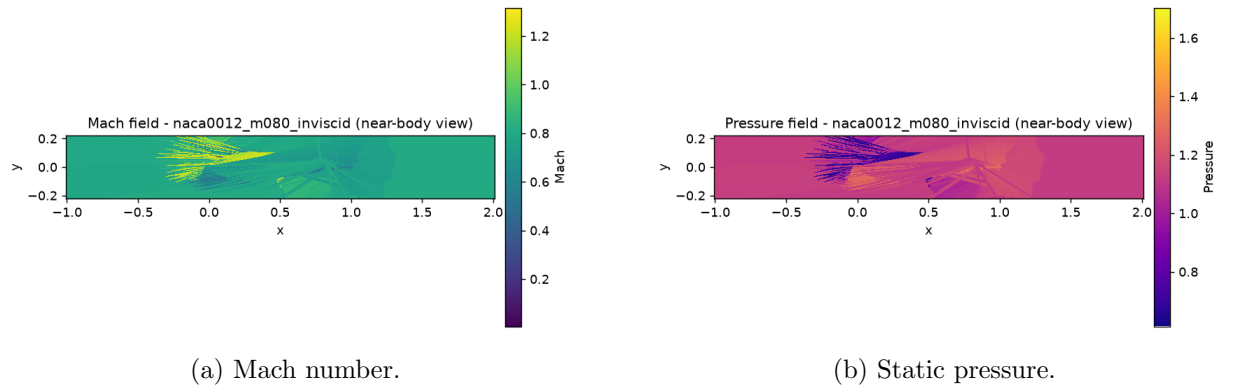
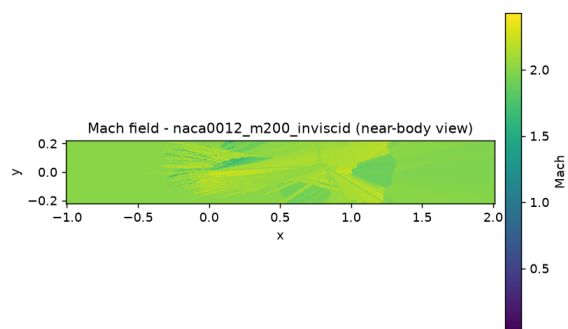
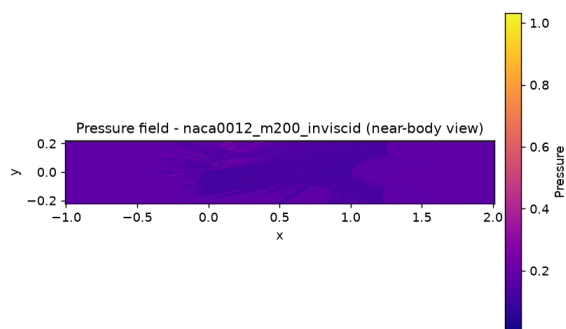


Figure 7: Near-body Mach and pressure contours for the Mach-0.8 inviscid airfoil, showing the shock structure.

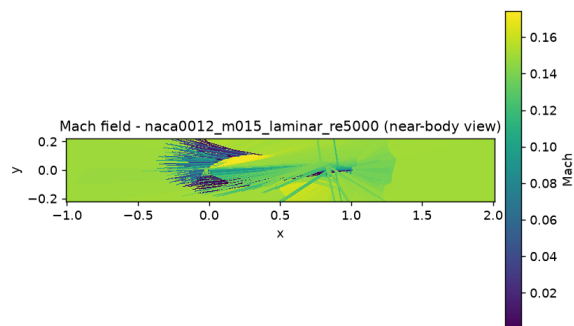


(a) Mach number.

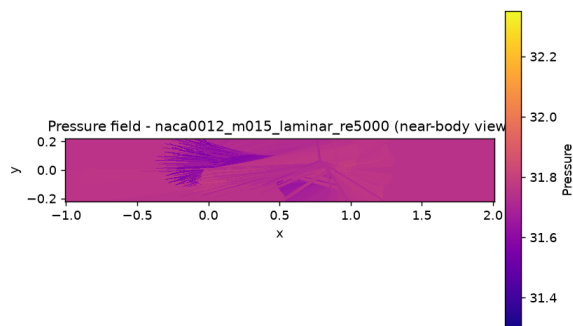


(b) Static pressure.

Figure 8: Near-body Mach and pressure contours for the Mach-2.0 inviscid airfoil.

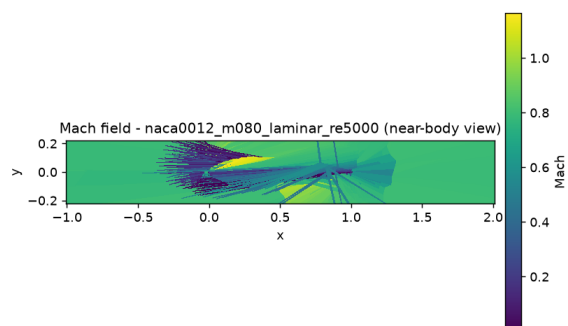


(a) Mach number.

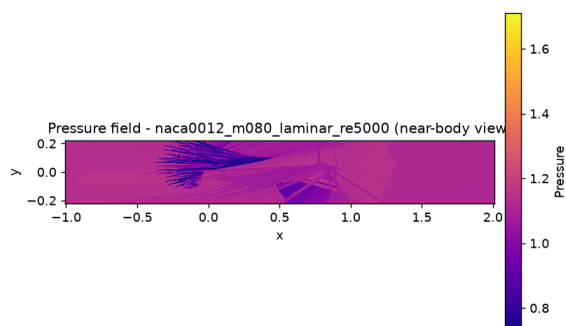


(b) Static pressure.

Figure 9: Mach-0.15 laminar NACA0012 near-body fields.



(a) Mach number.



(b) Static pressure.

Figure 10: Mach-0.8 laminar NACA0012 near-body fields.

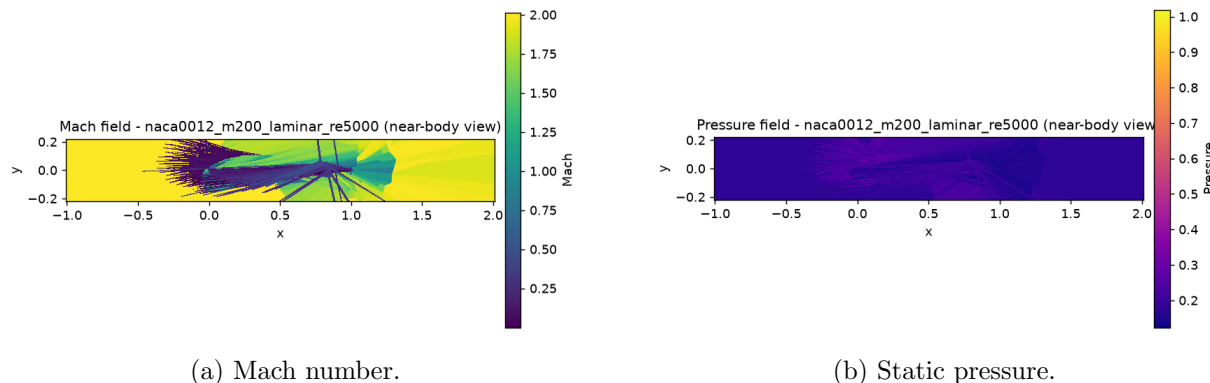


Figure 11: Mach-2.0 laminar NACA0012 near-body fields.

below also shows strong rank-count sensitivity: an $np=4$ run is near-zero-lift, while $np=8$ selects an asymmetric state. The result is therefore interpreted as insufficient wake resolution interacting with mesh asymmetry and partition-dependent roundoff/limiter paths, not as a clean validation case. The convergence and force histories still demonstrate the numerical solver's stability and reproducibility.

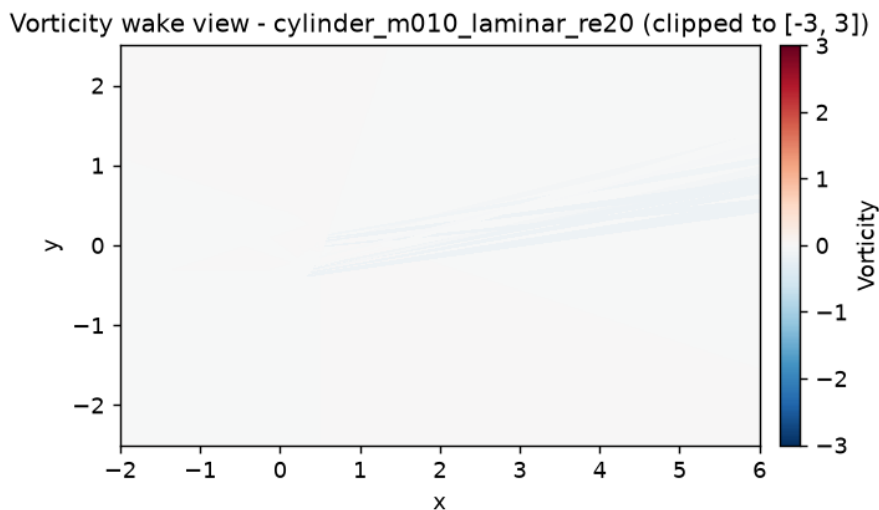
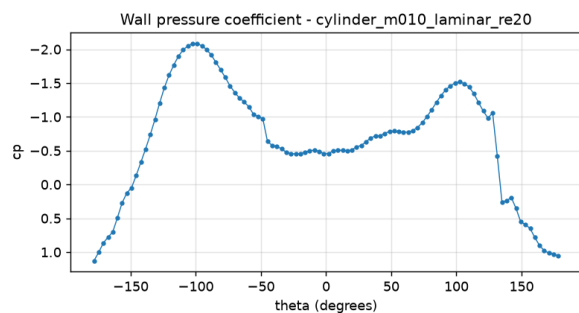


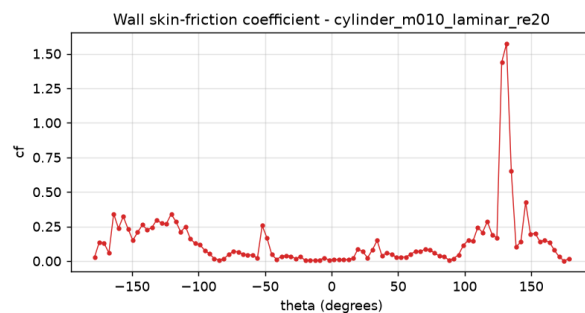
Figure 12: Reynolds-20 cylinder wake vorticity, clipped to $[-3, 3]$.

9.4 Cylinder at Reynolds 200

The Reynolds-200 case is a true BDF2 physical-time calculation. Force rows are written every physical step and intermediate field files are written every nondimensional time unit. fig. 16 shows the residual and force histories; fig. 18 shows the post-transient lift history and wake vorticity; fig. 19 shows near-body Mach and pressure. Figure 17 shows wall pressure and friction. The dominant frequency and Strouhal number are obtained by an unwrapped FFT of $C_L(t)$ over the last 65% of the full history ($t \geq 105$). Because $U_\infty = L_{\text{ref}} = 1$, the frequency in nondimensional time units equals the Strouhal number. The tabulated drag mean, lift rms, and peak amplitude

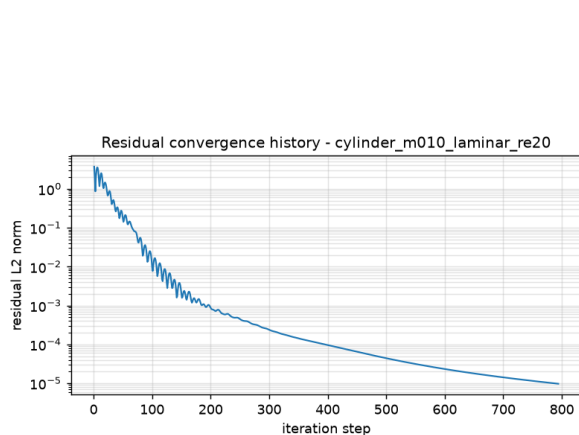


(a) Wall pressure coefficient.

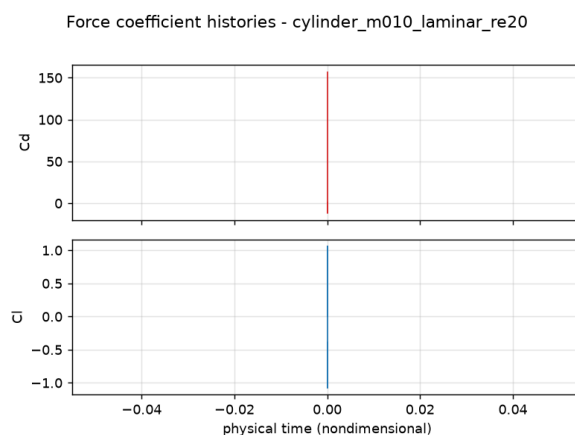


(b) Wall skin-friction coefficient.

Figure 13: Cylinder boundary pressure and friction at Reynolds 20 versus angle.

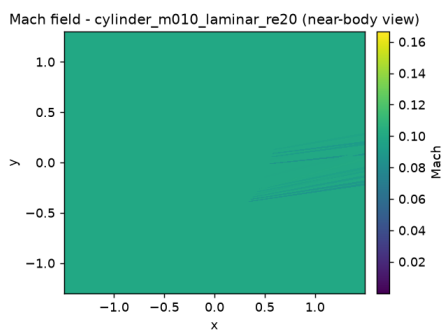


(a) Residual L2.

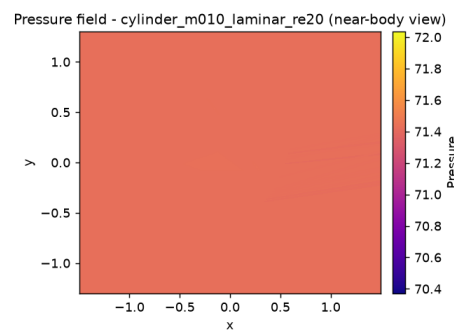


(b) C_L and C_D .

Figure 14: Reynolds-20 cylinder histories from `residuals.csv` and `forces.csv`.



(a) Mach number.



(b) Static pressure.

Figure 15: Reynolds-20 cylinder near-body Mach and pressure fields.

use the post-transient samples with $t \geq 100$. The resulting mean drag is $C_D \simeq 1.126$, the lift rms is about 0.027, and the dominant nondimensional frequency is about 0.113. This Strouhal number is below the canonical two-dimensional value near 0.2, consistent with the coarse wake resolution of the supplied mesh and the compressible low-Mach formulation. table 7 reports the quantitative statistics.

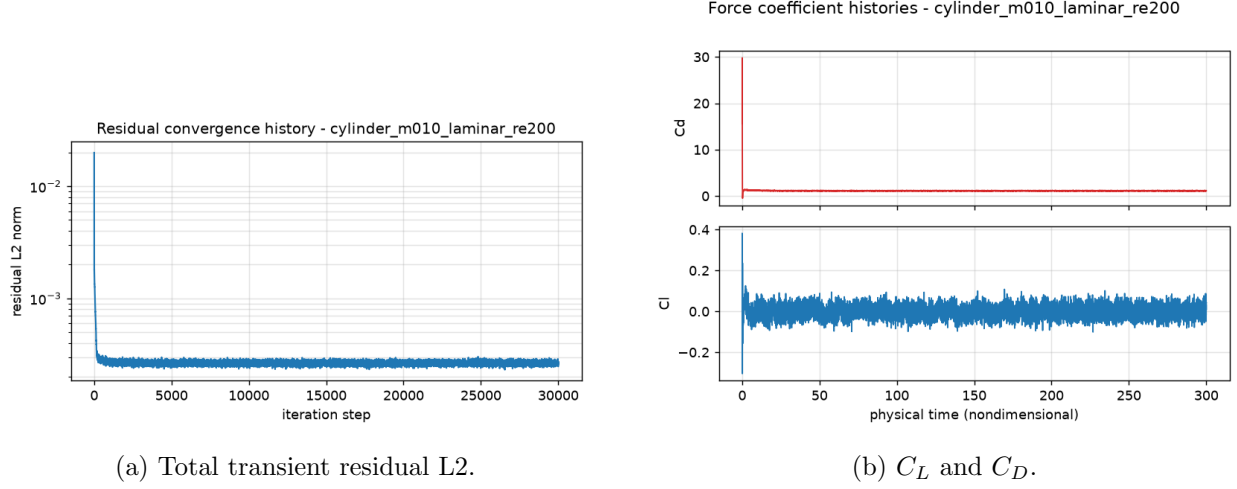


Figure 16: Reynolds-200 residual and force histories. Residuals are plotted against physical steps; forces are plotted against physical time.

Table 7: Reynolds-200 post-transient statistics from `forces.csv`.

Samples	$t_{\min}$	$\overline{C_D}$	$C_{L,rms}$	C_L peak amplitude	Frequency	St
20002	100.0	1.12646	0.02664	0.10461	0.11281	0.11281

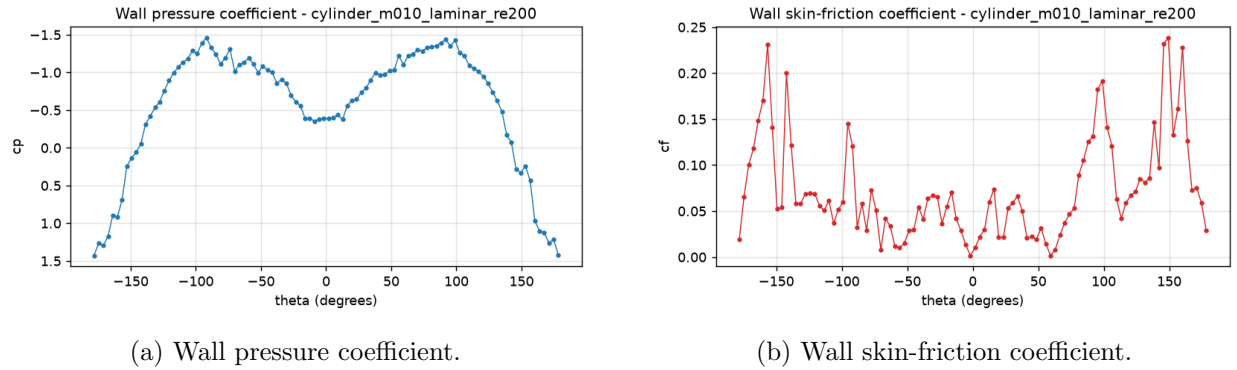
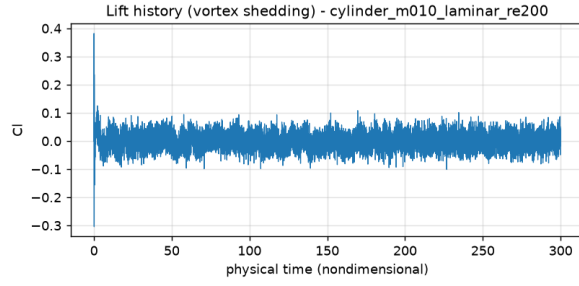


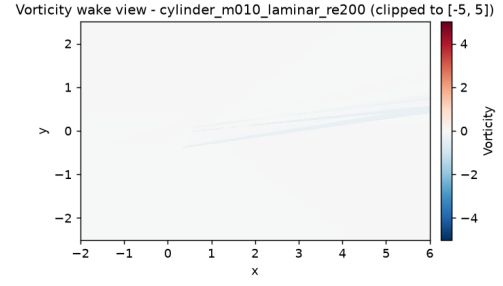
Figure 17: Cylinder boundary pressure and friction at Reynolds 200 versus angle.

10 Parallel Validation

table 8 compares the Mach-0.15 inviscid NACA0012 case at $np=2, 4$, and 8. The $np=8$ case is the production reference. All rank counts converged. The largest absolute drag difference is below

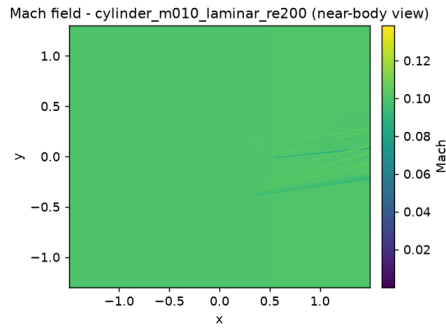


(a) Lift history.

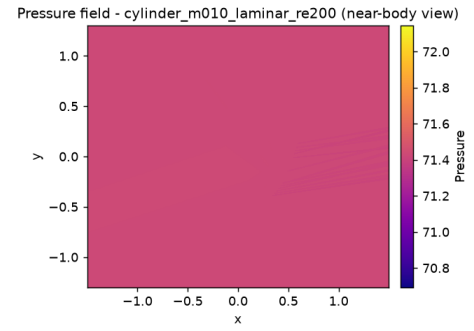


(b) Vorticity wake, clipped to $[-5, 5]$.

Figure 18: Reynolds-200 shedding diagnostics.



(a) Mach number.



(b) Static pressure.

Figure 19: Reynolds-200 near-body Mach and pressure fields.

6×10^{-4} ; because this inviscid force is itself only 6×10^{-3} , the relative difference is visible and is consistent with different partition-dependent roundoff and limiter-frozen paths. The corresponding lift differences are comparable in absolute size to the drag and are also attributable to these small asymmetric perturbations. The np=2, np=4, and np=8 wall times were 772, 888, and 349 s; the non-monotonic times reflect different final pseudo steps and partition-dependent communication volumes, not simple speedup. The machine-readable source is `report/rank_comparison.csv`.

Table 8: MPI rank-count comparison for the Mach-0.15 inviscid NACA0012 case.

Ranks	Final step	Mean C_D	ΔC_D vs np=8	Wall (s)
2	1186	0.005657	-0.000293	772
4	1111	0.005426	-0.000524	888
8	1034	0.005950	0	349

table 9 is the required cylinder rank comparison. Both rank counts converged to positive drag and a stable tail, but the np=4 run gives $C_D = 0.917$ with near-zero lift, whereas the np=8 reference gives $C_D = 0.847$ with $C_L = -0.390$. This large lift sensitivity is a warning that the Reynolds-20 cylinder problem has multiple weakly resolved numerical states; the comparison therefore supports the report’s decision not to claim clean physical validation for this case.

Table 9: MPI rank-count comparison for the Reynolds-20 cylinder case.

Ranks	Final step	Mean C_D	Mean C_L	Wall (s)
4	879	0.916972	0.012703	619
8	794	0.846906	-0.389676	176

11 Limitations and Failure Analysis

The principal limitations are:

1. The Reynolds-20 cylinder does not reproduce the canonical drag level and symmetric wake. The asymmetric supplied mesh and coarse wake resolution are contributors, and the rank-count comparison shows strong sensitivity to partition-dependent perturbations. A finer reflection-symmetric mesh and local refinement would be needed for validation.
2. The simple SGS relaxation requires conservative CFL caps and inner iteration counts; case-file CFL ramps up to 80–100 diverged during development with active reconstruction.
3. Frozen-coefficient inner iterations improve robustness but are less strict than a fully Newton-consistent Jacobian at every inner iteration.
4. The limiter can freeze for the remainder of a steady pseudo step if chatter is detected. The transient path refreshes the limiter every physical step, but very early transients still trigger the per-step freeze.
5. The Cartesian farfield domain is large but not a characteristic nonreflecting boundary for all unsteady waves.

6. The Rusanov transient test produced a steady two-vortex wake; the final Reynolds-200 run therefore uses the less dissipative Roe flux with an entropy fix. If shedding is still not statistically established, the run status reports that failure rather than inventing periodicity.

No failed or partial run is presented as final. Diagnostic directories are kept outside the submitted **results/** tree.